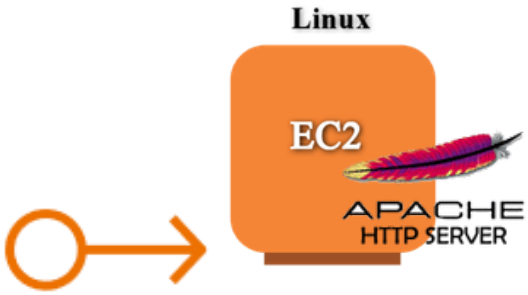
AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

Trainer: Aravindraj.G- Nminds Academy

AWS Academy Certified Educator

6.Configure Elastic IP Address to Windows Web Server in AWS



SRILEKA B

CSE-Artificial Intelligence & Machine Learning,

Sri Eshwar College of Engineering,

Coimbatore.

Objective

An Elastic IP (EIP) in AWS is a static, public IP address designed for dynamic cloud computing. It is associated with your AWS account and can be quickly associated or disassociated with any EC2 instance in your account. This feature is particularly useful when you need to maintain a consistent IP address for your resources, even when you stop and start EC2 instances.

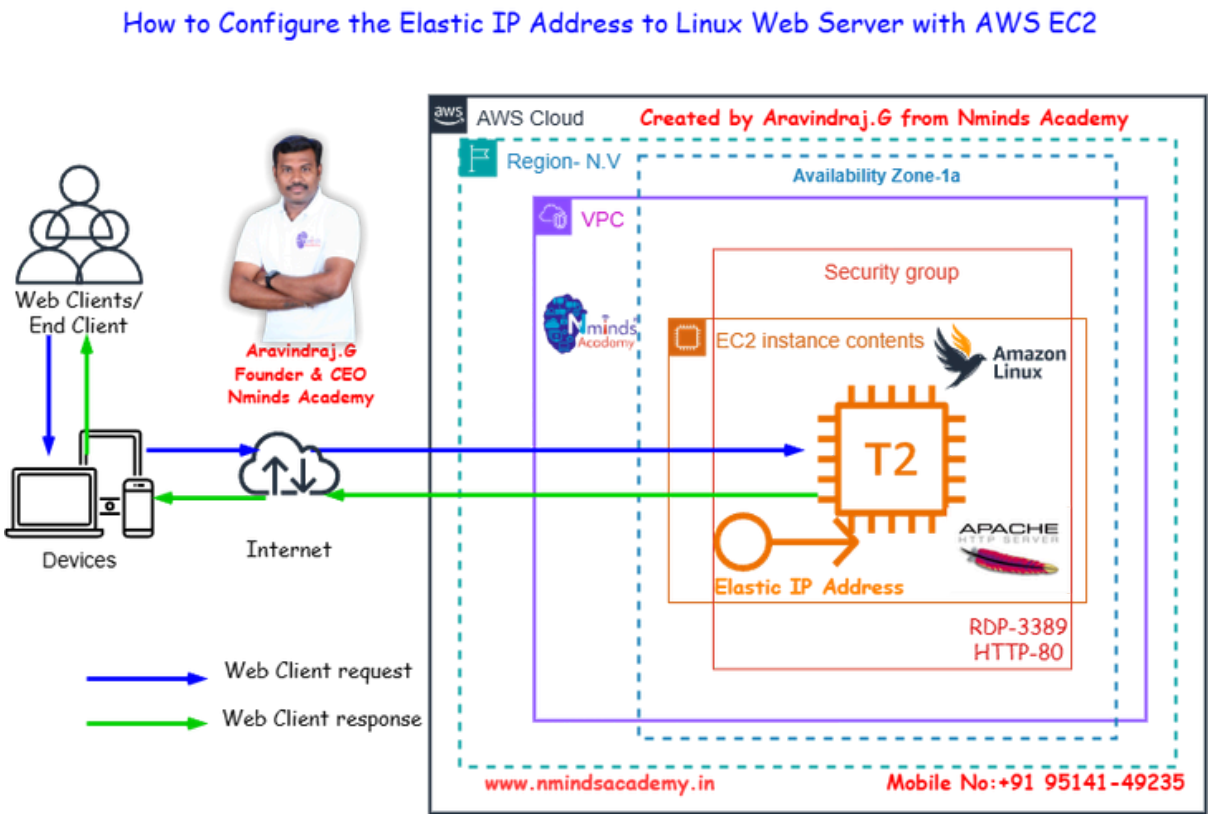
Common Use Cases for Elastic IP:

1. Highly Available Applications:
   - If you're running a service that requires high availability, you can use Elastic IPs to quickly reassign a static IP to a new instance if your primary instance fails, ensuring minimal downtime.
2. Web Servers:
   - If you host a website and need to ensure the IP address remains the same even if the underlying EC2 instance is restarted, an Elastic IP helps maintain this consistency.
3. Disaster Recovery:
   - Elastic IPs are useful for disaster recovery scenarios. If one instance goes down, you can quickly associate the EIP with a backup instance to ensure services are still accessible.

Best Practices:

- Use Elastic IPs only when necessary: Since they come with associated costs when unused, it's a good practice to release EIPs that are no longer required.
- Move IPs during instance failure: Instead of keeping an EIP permanently attached, use it as a failover method, reassociating it to a new instance when needed.
- Monitor EIP Usage: Periodically review your usage to ensure you're not paying for unnecessary Elastic IP addresses.

Topology



Used Services in AWS

Used Services in Windows Server

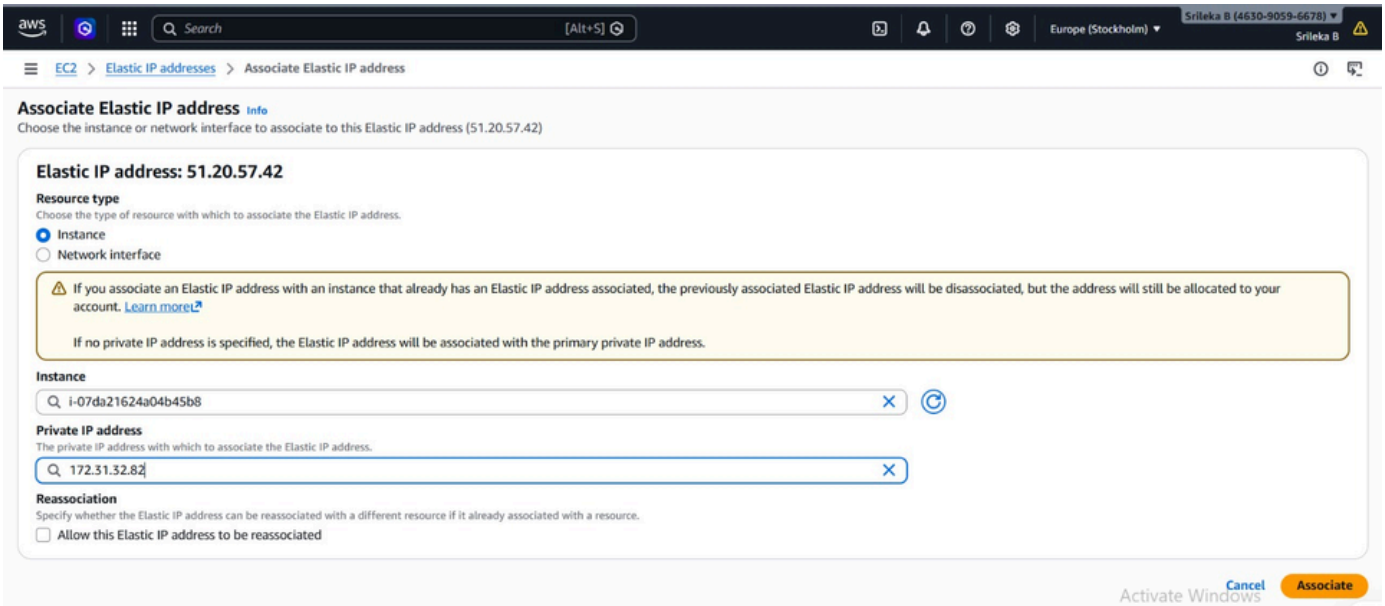
Amazon EC2

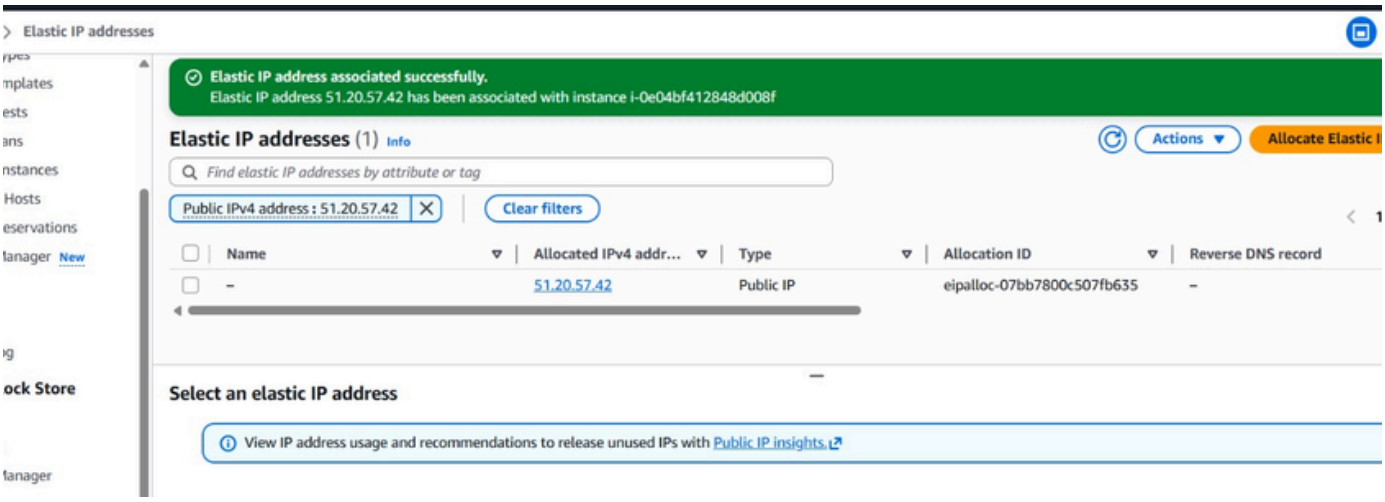
APACHE HTTP SERVER

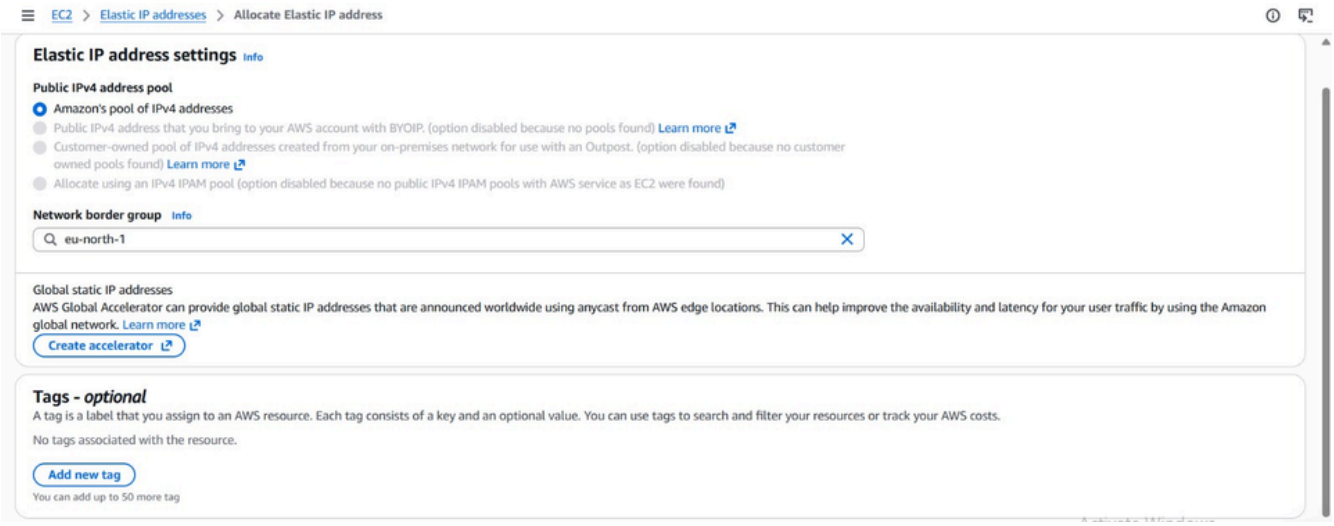
Apache Webserver

Amazon ElasticIPAddress

Execution Tasks:






EC2 > Elastic IP addresses > Allocate Elastic IP address
Elastic IP address settings Info
Public IPv4 address pool
Amazon's pool of IPv4 addresses
Public IPv4 address that you bring to your AWS account with BYOIP. (option disabled because no pools found) Learn more
Customer-owned pool of IPv4 addresses created from your on-premises network for use with an Outpost. (option disabled because no customer owned pools found) Learn more
Allocate using an IPv4 IPAM pool (option disabled because no public IPv4 IPAM pools with AWS service as EC2 were found)
Network border group Info
eu-north-1
Global static IP addresses
AWS Global Accelerator can provide global static IP addresses that are announced worldwide using anycast from AWS edge locations. This can help improve the availability and latency for your user traffic by using the Amazon global network. Learn more
Create accelerator
Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
No tags associated with the resource.
Add new tag
You can add up to 50 more tag